Fiscal Inaction as Monetary Support

by George-Marios Angeletos, Chen Lian, and Christian Wolf

Discussion by Marco Bassetto

Federal Reserve Bank of Minneapolis

This discussion reflects my views and not necessarily those of the Federal Reserve Bank of Minneapolis or the Federal Reserve System

What the paper does

- Bring systematic fiscal policy into HANK
- Analyze its optimal usage

Plan of Discussion

- Stabilization policy in RANK
- Stabilization policy in HANK
- More on interest rate smoothing
- Quantitative questions

Plain Vanilla RANK

$$y_t = E_t y_{t+1} - \sigma(r_t + v_t)$$

$$\pi_t = \beta E_t \pi_{t+1} + \kappa y_t + u_t$$

$$d_{t+1} = r_t d_t - t_t$$

Stabilizing demand shocks in RANK

$$y_t = E_t y_{t+1} - \sigma(r_t + v_t)$$

- Divine coincidence: just use r_t to offset v_t
- Will need to adjust lump-sum taxes, but Ricardian equivalence holds

Stabilizing demand shocks in RANK

$$y_t = E_t y_{t+1} - \sigma(r_t + v_t)$$

- Divine coincidence: just use r_t to offset v_t
- Will need to adjust lump-sum taxes, but Ricardian equivalence holds
- Problem with ZLB

Stabilizing demand shocks in RANK

$$y_t = E_t y_{t+1} - \sigma(r_t + v_t)$$

- Divine coincidence: just use r_t to offset v_t
- Will need to adjust lump-sum taxes, but Ricardian equivalence holds
- Problem with ZLB
- Can use gov't spending (part of v_t), but need shovel-ready projects
- Taxes and transfers are useless, except for "helicopter money:" Ricardian equivalence

Stabilizing supply shocks in RANK

$$y_t = E_t y_{t+1} - \sigma(r_t + v_t)$$
$$\pi_t = \beta E_t \pi_{t+1} + \kappa y_t + u_t$$

- Divine coincidence breaks down
- Still the job of interest rate policy, trading off output vs. inflation
- Can spread the pain over time (under commitment)
- Fiscal policy subject to the same problem as interest rate policy

HANK economy: what changes?

- New intertemporal IS:

$$y_t = E_t y_{t+1} - \sigma(r_t - r_t^* + v_t)$$

$$r_t^* = \frac{(1-\beta\omega)(1-\omega)}{\sigma\omega} E_t d_{t+1}$$

- "Natural rate of interest" depends on debt
- Monetary policy has to offset movements in r^*
- Divine coincidence still holds
- Now, can use tax/transfer policy: $t_t \downarrow \Longrightarrow d_{t+1} \uparrow \Longrightarrow r_t^* \uparrow \Longrightarrow y_t \uparrow$

Tax policy as an automatic stabilizer

- Suppose

$$t_t = \tau_y y_t + \tau_d d_t + \beta \frac{D^{ss}}{Y^{ss}} r_t$$

- $v_t \downarrow \Longrightarrow y_t \downarrow \Longrightarrow t_t \downarrow$
- Do not have divine coincidence for automatic stabilizers: need $y \downarrow$

Tax policy as an automatic stabilizer

- Suppose

$$t_t = \tau_y y_t + \tau_d d_t + \beta \frac{D^{ss}}{Y^{ss}} r_t$$

- $v_t \downarrow \Longrightarrow y_t \downarrow \Longrightarrow t_t \downarrow$
- Do not have divine coincidence for automatic stabilizers: need $y \downarrow$
- Use the power of forward guidance:
  - If $E_t d_{t+2} > 0$, get $r^*_{t+1} \uparrow \Longrightarrow E_t y_{t+1} \uparrow \Longrightarrow y_t \uparrow$
  - Set τ_d small, so debt keeps on giving!

Supply shocks: tax policy as an automatic de-stabilizer

- Contractionary supply shock calls for $r_t \uparrow$ (depressing output to reduce inflation)
- Increasing r_t^* through t_t goes the wrong way

Supply shocks: tax policy as an automatic de-stabilizer

- Contractionary supply shock calls for $r_t \uparrow$ (depressing output to reduce inflation)
- Increasing r_t^* through t_t goes the wrong way
- Now the question becomes quantitative ($+$ distortionary taxes)

Do we need fiscal policy?

- Standard answer: no, we already have monetary policy...
- Except at the ZLB
- What if moving interest rates is costly?
- Then fiscal policy is handy

Is moving interest rates costly? 1. In levels

- Paper baseline: level of interest rates

Is moving interest rates costly? 1. In levels

- Paper baseline: level of interest rates
- Can appeal to the Friedman rule (should have i=0)

Is moving interest rates costly? 1. In levels

- Paper baseline: level of interest rates
- Can appeal to the Friedman rule (should have i=0)
- ZLB (should have $i >> 0$)

Is moving interest rates costly? 1. In levels

- Paper baseline: level of interest rates
- Can appeal to the Friedman rule (should have i=0)
- ZLB (should have $i >> 0$)
- Humphrey-Hawkins: "maximum employment, stable prices, and moderate long-term interest rates."

  Judd and Rudebusch (1999) The goals of "stable prices" and "moderate long-term interest rate" are related because nominal interest rates are boosted by a premium over real rates equal to expected future inflation. Thus, "stable prices" will typically produce long-term interest rates that are "moderate."

Is moving interest rates costly? 2. In differences

- Woodford (1999): if it is costly in levels, then there is a role for smoothing (if CB acts under discretion, a smoothing objective is good)
- Tealbook simulations include smoothing (under discretion)

Quantitative question #1

- Optimal policy: half life of debt of about 10 years
- Slightly faster than the Maastricht rules ($.95^{10} \approx .6$)

Quantitative question #1

- Optimal policy: half life of debt of about 10 years
- Slightly faster than the Maastricht rules ($.95^{10} \approx .6$)
- What would it be in RANK with distortionary taxes?

Magnitude of effects

- CBO estimate: an extra 1% in debt/GDP $\Longrightarrow$ 2bps on r^*
- CBO estimate of debt impact from automatic stabilizers:
  - 10% of GDP (20bps) for the Great Recession (most of it after the recession was over, slow recovery)
  - 2% of GDP for COVID (4bps)
- Very long lasting, effect magnified.